Software Requirements Specification: Department Tower Clash

1. Introduction

1.1 Purpose

This document specifies the requirements of **Department Tower Clash**, a desktop tower-vs-tower battle game built in Java. It is written for the development team (Group 8), the course instructor, and anyone who will test, maintain or extend the system.

1.2 Scope

Department Tower Clash is a single-player and two-player game in which each player controls a tower themed after a school department and sends units to destroy the opposing tower. The system covers:

- Player accounts (register, log in, profile, change password, and a security question to reset a forgotten password) with progress saved per player
- A dashboard with personal statistics and a top-10 leaderboard
- Department, lineup and stage selection
- Real-time battles against an AI opponent (five story stages and an Endless mode)
- Permanent tower upgrades bought with coins
- LAN 1v1 battles between two computers on the same network
- Match history and a MySQL database, with automatic offline saving when the database is not available

Out of scope: online play over the internet, mobile versions, in-app purchases, and an administrator panel (planned as future work).

1.3 Definitions, acronyms and abbreviations

Term	Meaning
Department	One of three playable sides: Hospitality Management (red), Education (blue), Computer Science (yellow)
Tower	A player's base. The match is lost when its HP reaches 0
Unit	A character spawned by a player that walks toward and attacks the enemy
Lineup	The 3 units (out of 5 per department) a player brings into a battle
Energy	The resource that fills over time and is spent to spawn units and upgrade the tower
Ultimate	A once-charged, department-specific special skill
HP	Hit points (health)

Term	Meaning
AI	The computer-controlled opponent
LAN	Local Area Network: computers on the same Wi-Fi or wired network
DBMS	Database Management System (MySQL / MariaDB through XAMPP)
JDBC	Java Database Connectivity, the Java API used to talk to MySQL
SHA-256	A one-way hash function used to store passwords
FR / NFR	Functional requirement / non-functional requirement

1.4 References

- IEEE Std 830-1998, *Recommended Practice for Software Requirements Specifications*
- Project files: `LAN_SETUP.txt` and `sql/tower_battle_db.sql` in the TowerBattle project folder

1.5 Overview

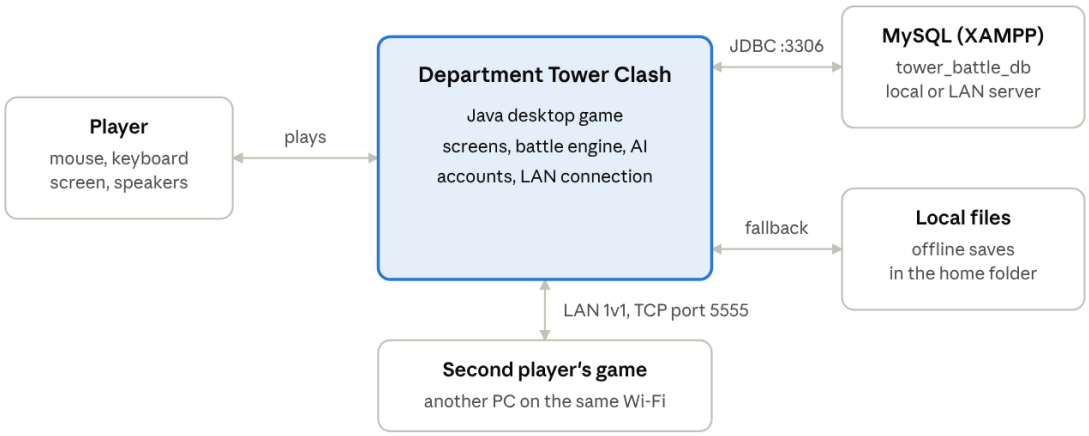
Section 2 describes the product as a whole. Section 3 lists the external interfaces. Section 4 lists the functional requirements by feature, Section 5 the non-functional requirements, and Section 6 the data the system stores. Section 7 summarizes the use cases.

2. Overall Description

2.1 Product perspective

Department Tower Clash is a new, self-contained desktop application. It runs as one Java program on the player's computer and connects to two optional outside parts: a MySQL database (local or on another computer in the LAN) and, for LAN 1v1, the game running on a second computer. When MySQL cannot be reached, the game keeps working and saves to files on the same computer.

The game runs on its own; MySQL and a second computer are optional



system context · the game and the 4 parts it talks to

The player uses only the game; it saves to MySQL when it can reach it and to local files when it cannot.

2.2 Product functions

1. Create an account, log in and log out
2. View a dashboard with personal statistics and a leaderboard
3. View a profile and the match history, change the password, and reset a forgotten password by answering a security question
4. Choose a department, pick a lineup of 3 units, and choose a stage
5. Play a real-time battle against the AI: spawn units, upgrade the tower, use the Ultimate
6. Earn coins and stars, unlock stages, and buy permanent tower upgrades
7. Play Endless mode for the longest survival time
8. Host or join a LAN 1v1 battle against another player
9. Choose where data is saved (this computer or a MySQL server) and adjust game options

2.3 User classes and characteristics

User class	Description	Technical skill
Player	A student or casual gamer who registers, plays and checks the leaderboard	Basic computer use
LAN host	A player who starts a LAN 1v1 match and shares their IP address	Basic; can read an IP address
Server operator	The person (e.g. the teacher or a group member) who runs XAMPP so that a class shares one leaderboard	Can start XAMPP and run a SQL script

2.4 Operating environment

- **Operating system:** Windows 10 or 11 (any OS with a matching Java runtime should also work)
- **Runtime:** Java Development Kit / Runtime 26, the version the project is compiled for
- **Development tool:** Apache NetBeans with the Ant build
- **Database:** MySQL (MariaDB) through XAMPP, default port 3306
- **Library:** MySQL Connector/J (included in the project's `lib` folder)
- **Network (LAN 1v1 only):** both computers on the same Wi-Fi or LAN; TCP port 5555 allowed through the firewall

2.5 Design and implementation constraints

- Written in Java SE only (Swing, Java2D, Java Sound, JDBC, Sockets); no game engine.
- All graphics and sounds are generated by code; the project contains no image or audio files.
- The game draws on a fixed 320 x 180 virtual screen, scaled to fit the window.
- LAN 1v1 requires both computers to run the same version of the game.
- Course requirement: developed in NetBeans as a school project.

2.6 Assumptions and dependencies

- XAMPP's MySQL is started before the game when database saving is wanted; otherwise the game saves locally.
- The default XAMPP account (`root`, no password) is used on a single computer. A shared LAN server uses the `towergame` account created by `sql/lan_setup.sql`.
- School networks may block traffic between computers; a phone hotspot can be used instead for LAN features.

3. External Interface Requirements

3.1 User interfaces

All screens share a pixel-art style with wooden-board panels, large readable text, and controls that work with both mouse and keyboard (arrow keys, Enter, Tab, Esc).

Screen	Purpose	Main controls
Login	Sign in to an account	Username, Password, LOG IN, REGISTER, FORGOT PASSWORD?, DATABASE, EXIT
Register	Create a new account	Step 1: Username, Email, Password, Confirm, NEXT. Step 2: Security question (click to change), Answer, CREATE ACCOUNT, BACK
Forgot Password	Reset a forgotten password with the security question	Step 1: Username or Email, NEXT. Step 2: Answer, New Password, Confirm New, RESET PASSWORD, BACK
Database Settings	Choose where accounts are saved	Server IP, Port, MySQL User, Password, CONNECT, THIS COMPUTER, BACK
Dashboard	Home screen after login	Your Stats, Leaderboard tabs, PROFILE, HISTORY, PLAY, LAN 1V1, UPGRADES, HELP, OPTIONS, LOG OUT
Profile	Account details and progress summary	HISTORY, CHANGE PASSWORD, SECURITY QUESTION, BACK
Match History	List of finished matches, 8 per page	PREV, NEXT, BACK
Change Password	Replace the current password	Current, New, Confirm New, SAVE PASSWORD, BACK
Security Question	Set or change the security question and answer	Question (click to change), Answer, Password (when changing), SAVE, BACK or LATER
Tower Select	Choose a department	Three tower cards, NEXT, BACK

Screen	Purpose	Main controls
Lineup	Pick 3 of 5 units	Unit cards, RESET, NEXT, BACK
Stage Select	Choose a stage or Endless	Stage cards, ENDLESS, FIGHT!, BACK
Battle	Play the match	Unit buttons (1, 2, 3), Tower upgrade (U), Ultimate (R), Speed x1/x2 (F), Pause (P / Esc)
Upgrades	Buy permanent tower levels	Upgrade buttons per department
LAN 1v1 Lobby	Host or join a player-vs-player match	Department buttons, CHANGE LINEUP, HOST GAME, Host IP field, JOIN GAME, CANCEL
How to Play, Options, Credits	Help, settings, team credits	Toggles for sound, music, fullscreen, pixel-perfect and smooth text

3.2 Hardware interfaces

- Keyboard and mouse (or touchpad)
- Display of at least 640 x 360 pixels; the game scales to larger screens and supports fullscreen
- Speakers or headphones for music and sound effects (optional)
- Network adapter (Wi-Fi or Ethernet) for a remote database or LAN 1v1

3.3 Software interfaces

Interface	Used for	Details
MySQL / MariaDB (XAMPP)	Accounts, progress, match history, leaderboard	Database `tower_battle_db`, port 3306; tables are created automatically on first connection
MySQL Connector/J (JDBC driver)	Connecting Java to MySQL	JAR in the project's `lib` folder
Local file system	Offline saving and settings	Folder `.department-tower-clash` in the user's home folder
phpMyAdmin (optional)	Viewing the database tables	Opened through XAMPP's Apache

3.4 Communications interfaces

- **Database:** JDBC over TCP to the MySQL server (default `localhost:3306`); connection timeout of 3 seconds.
- **LAN 1v1:** a direct TCP connection between the two computers on port 5555. The host listens; the second player connects using the host's IP address. Both games exchange a handshake (game version, player name, department, tower level, lineup), then only player actions and periodic check values during the match.

4. Functional Requirements

Each requirement uses "shall" and has an ID for testing and traceability. Priority: **High** = needed for the game to work; **Medium** = important but the game works without it.

4.1 Account management

ID	Requirement	Priority
FR-01	The system shall let a new player register in two steps; step 1 asks for a username, email and password typed twice.	High
FR-02	The system shall accept usernames of 3 to 16 characters using only letters, numbers and underscore.	High
FR-03	The system shall accept passwords of 6 to 32 characters with at least one letter and one number and no spaces.	High
FR-04	The system shall reject a username or email that is already used, and show the reason before moving to step 2.	High
FR-05	In step 2, the player shall choose one of 7 security questions and type an answer (2 to 40 characters) before the account is created.	High
FR-06	The system shall log a player in only when the username and password match, and show "WRONG USERNAME OR PASSWORD" otherwise.	High
FR-07	The system shall load the player's saved progress after login and save it after every match and purchase.	High
FR-08	The system shall let the player log out, and remember the last username on the Login screen.	Medium
FR-09	The system shall import an old single-player save file into the first account created on that computer.	Medium

4.2 Dashboard and leaderboard

ID	Requirement	Priority
FR-10	The system shall show a dashboard after login with matches played, wins and losses, win rate, high score, fastest win, stars, LAN 1v1 record, best Endless time and coins.	High
FR-11	The system shall show a top-10 leaderboard with four tabs: high score, wins, stars, and Endless time.	High
FR-12	The system shall highlight the logged-in player's row and show their rank even when they are outside the top 10.	Medium

ID	Requirement	Priority
FR-13	The system shall give players with the same score the same rank.	Medium

4.3 Profile, match history, password and security question

ID	Requirement	Priority
FR-14	The system shall show a profile with username, email, player ID, date joined, number of saved matches, progress summary and tower levels.	Medium
FR-15	The system shall list the player's finished matches newest first, 8 per page, with date, mode, both departments, result, score and time.	Medium
FR-16	The system shall show damage dealt, units defeated, tower HP left and coins earned for the match the player points at.	Medium
FR-17	The system shall let a logged-in player change their password only after entering the current password correctly.	High
FR-18	The system shall apply the same password rules as registration and reject a new password equal to the old one.	High
FR-19	The player shall change their security question and answer from the Profile after entering the current password.	Medium
FR-20	The system shall ask players whose account has no security question (accounts made before this feature) to set one right after login, with a LATER option.	Medium

4.4 Forgot password

ID	Requirement	Priority
FR-21	The Login screen shall have a FORGOT PASSWORD? button; the player types a username or email and the system shows that account's security question.	High
FR-22	When the answer is correct, the system shall save the new password (same rules as registration, typed twice) and return to Login with the username filled in.	High
FR-23	The system shall compare answers without case and extra spaces ("San Pablo" equals "san pablo").	Medium
FR-24	After 5 wrong answers in a row, the system shall block password reset for that account for 15 minutes; a successful reset clears the counter.	High

4.5 Department, lineup and stage selection

ID	Requirement	Priority
FR-25	The system shall offer three departments: Hospitality Management, Education and Computer Science, each with its own tower, tower attack and Ultimate.	High
FR-26	The system shall give each department 5 units and let the player choose exactly 3 for the lineup.	High
FR-27	The system shall save each department's lineup per player and offer a RESET to the default lineup.	Medium
FR-28	The system shall show each unit's role, cost, HP, damage, range, speed and ability before it is chosen.	Medium
FR-29	The system shall show five story stages and lock each stage until the one before it is won.	High
FR-30	The AI opponent shall use a different department from the player, chosen at random, with a random lineup.	High

4.6 Battle

ID	Requirement	Priority
FR-31	The system shall start each side with 40 energy, refill it over time up to 100, and add a bonus for each enemy unit defeated.	High
FR-32	The player shall spawn a lineup unit by clicking its button or pressing 1, 2 or 3 when they have enough energy.	High
FR-33	Units shall walk toward the enemy, attack enemy units and the enemy tower in range, and use their ability.	High
FR-34	The system shall apply role counters: FAST beats RANGED, RANGED beats TANK, TANK beats FAST.	High
FR-35	The player shall upgrade the tower during a match (up to level 3) by spending energy.	High
FR-36	Each tower shall have an Ultimate whose meter fills over time and when dealing damage; the player uses it with R or its button when full.	High
FR-37	The system shall start Overtime in long matches: double energy and towers taking increasing damage, so every match ends.	Medium
FR-38	The player shall pause the match (P or Esc) and then resume or quit, retry from the result screen, and switch game speed between x1 and x2	Medium

ID	Requirement	Priority
	(F).	
FR-39	The match shall end when a tower's HP reaches 0 and show the result, score, stars and coins earned.	High

4.7 Rewards, progress and Endless mode

ID	Requirement	Priority
FR-40	The system shall award up to 3 stars per stage: 1 for winning, 1 for keeping 50% tower HP, 1 for winning in 2:00 or less.	Medium
FR-41	The system shall award coins after every match, more for a win.	High
FR-42	The player shall spend coins on permanent tower levels (up to level 5) for each department.	Medium
FR-43	The system shall unlock Endless mode after Stage 3 is cleared and record the longest survival time and best score.	Medium
FR-44	The system shall save every finished match to the match history.	High

4.8 LAN 1v1

ID	Requirement	Priority
FR-45	A player shall host a LAN match and see their computer's IP address.	High
FR-46	A second player shall join by typing the host's IP address.	High
FR-47	The system shall refuse to start when the two games have different versions.	High
FR-48	Both computers shall show the same battle: the host plays the left tower and the joiner the right tower, with "YOU" marking the player's own tower.	High
FR-49	A player shall surrender with P or Esc; the match shall have no pause or speed-up.	Medium
FR-50	The system shall end the match without a result if no message arrives for 10 seconds.	High
FR-51	The system shall record LAN wins and losses on each player's account.	Medium

4.9 Database settings and options

ID	Requirement	Priority

ID	Requirement	Priority
FR-52	The system shall connect to MySQL at start-up, creating the database and tables (and adding any missing columns) if they do not exist.	High
FR-53	The system shall save to local files automatically when MySQL cannot be reached, and say where data is being saved.	High
FR-54	The player shall set the server IP, port, MySQL user and password, test the connection, and return to this computer's settings with one button.	Medium
FR-55	The system shall remember the database settings for the next start.	Medium
FR-56	The player shall turn sound, music, fullscreen, pixel-perfect scaling and smooth text on or off.	Medium
FR-57	The system shall provide a How to Play screen and a Credits screen.	Medium

5. Non-Functional Requirements

5.1 Performance

ID	Requirement
NFR-01	Battles shall run at about 60 frames per second on a typical school laptop; menus at about 30.
NFR-02	A database operation (login, save, leaderboard) shall finish within 3 seconds on a working connection; an unreachable server shall be detected within 3 seconds.
NFR-03	A LAN 1v1 action shall take effect on both computers 0.1 seconds after it is pressed.
NFR-04	The game shall start and reach the Login screen within 5 seconds.

5.2 Security

ID	Requirement
NFR-05	Passwords shall never be stored in plain text; the system shall store a SHA-256 hash with a random 16-byte salt per account.
NFR-06	Changing or resetting a password shall generate a new salt.
NFR-07	Security answers shall be stored the same way as passwords: only a salted SHA-256 hash of the answer, never the answer itself.
NFR-08	Password reset by security question shall be limited to 5 wrong answers, then locked for 15 minutes, to stop guessing.
NFR-09	Every SQL statement that uses player input shall be a PreparedStatement to

ID	Requirement
	prevent SQL injection.
NFR-10	Login errors shall not reveal whether the username or the password was wrong.
NFR-11	A shared LAN database shall use a limited MySQL account (`towergame`) that can access only `tower_battle_db`.
NFR-12	Password fields shall hide typed characters.

5.3 Usability

ID	Requirement
NFR-13	Every screen shall be usable with the mouse alone and with the keyboard alone.
NFR-14	Text shall stay readable at every window size, with an option for smooth (non-pixel) letters.
NFR-15	Error messages shall say what is wrong in plain words (for example "MYSQL IS NOT RUNNING (START IT IN XAMPP)").
NFR-16	A first-time player shall see in-battle tips during their first two matches.
NFR-17	Resetting a forgotten password shall need no email, internet or administrator.

5.4 Reliability and availability

ID	Requirement
NFR-18	The game shall keep working without MySQL by saving to local files.
NFR-19	A broken line in a local history file shall be skipped without losing the rest of the history.
NFR-20	Both LAN games shall compare a game-state fingerprint every second to detect if they fall out of step.
NFR-21	A battle shall pause automatically when the game window loses focus (AI battles only).
NFR-22	A database made by an older version of the game shall be upgraded automatically (missing columns added) without losing accounts.

5.5 Portability

ID	Requirement
NFR-23	The game shall run on any computer with a matching Java runtime, from NetBeans

ID	Requirement
	or as a JAR with its `lib` folder.
NFR-24	The game shall adapt to screen sizes and Windows display scaling (100% to 200%).

5.6 Maintainability

ID	Requirement
NFR-25	The code shall be organized into packages: `app`, `ui`, `game`, `model`, `db`, `net`, `audio`.
NFR-26	Storage shall go through one `AccountStore` interface so that MySQL and local saving can be swapped without changing the rest of the game.
NFR-27	A new unit, stage or department shall be addable by writing one class or one line, without changing the battle engine.

6. Data Requirements

All data lives in the MySQL database `tower_battle_db` (5 tables, InnoDB, utf8mb4). The game creates the database and tables itself; `sql/tower_battle_db.sql` holds the same script for manual setup. Deleting a user also deletes their progress and matches (`ON DELETE CASCADE`).

6.1 Tables

Table	Purpose	Key fields
`users`	One row per account, with its security question	`user_id` (PK, auto-increment), `username` (unique), `email` (unique), `password_hash` (SHA-256, 64 hex characters), `salt` (Base64), `created_at`, `last_login`, `security_question`, `security_answer_hash` (SHA-256 of the answer), `security_salt`, `reset_attempts` (wrong answers in a row), `reset_locked_until` (milliseconds, 0 = not locked)
`player_progress`	One row per account with the numbers the leaderboard sorts by	`user_id` (PK, FK to users), `coins`, `unlocked_stage`, `matches_played`, `wins`, `losses`, `high_score`, `best_win_seconds`, `total_stars`, `best_endless_seconds`, `best_endless_score`, `save_data` (tower levels, lineups, stars per stage, LAN record), `updated_at`
`match_history`	One row per finished match	`match_id` (PK), `user_id` (FK), `stage_number` (FK), `player_tower` (FK), `enemy_tower` (FK), `result` (WIN / LOSS), `score`, `duration_seconds`, `damage_dealt`, `units_defeated`, `tower_hp_left`

Table	Purpose	Key fields
		(%), coins_earned, played_at
towers	The 3 departments (lookup)	tower_code (PK: HOSPITALITY, EDUCATION, COMPSCI), tower_name, department, short_name, color_hex
stages	The stages and modes (lookup)	stage_number (PK: 1 to 5 story, 6 Endless, 7 LAN 1v1), stage_name, description, difficulty, ai_multiplier, reward_coins

6.2 Relationships

- One users row has exactly one player_progress row (1 : 1).
- One users row has many match_history rows (1 : N).
- Each match_history row refers to one stages row and two towers rows (the player's and the enemy's).

6.3 Local (offline) storage

When MySQL is not available, the same data is kept in the folder .department-tower-clash/accounts in the user's home folder:

File	Contents
<username>.account	User ID, username, email, salt, password hash, date created, security question, answer hash, wrong-answer counter
<username>.save	The player's progress
<username>.history	One line per finished match
../database.properties	The last database settings (host, port, user, password)

7. Use Cases

The system has 16 use cases. The Player is the main actor; the MySQL server and the second LAN player are supporting actors.

ID	Use case	Actor	Main steps	Requirements
UC-01	Register	Player	Step 1: username, email, password twice → NEXT. Step 2: pick a security question, type the answer → CREATE ACCOUNT → Dashboard	FR-01 to FR-05, FR-09
UC-02	Log in	Player	Type username and password → LOG IN → progress loads → Dashboard (or Security	FR-06, FR-07, FR-20

ID	Use case	Actor	Main steps	Requirements
			Question if none is set)	
UC-03	Log out	Player	LOG OUT → confirm → Login screen	FR-08
UC-04	View dashboard and leaderboard	Player	Open Dashboard → switch tabs	FR-10 to FR-13
UC-05	View profile	Player	Dashboard → PROFILE	FR-14
UC-06	View match history	Player	HISTORY → page with PREV / NEXT → point at a row for details	FR-15, FR-16
UC-07	Change password	Player	Profile → CHANGE PASSWORD → current, new, confirm → SAVE	FR-17, FR-18
UC-08	Set or change security question	Player	After login (if none) or Profile → SECURITY QUESTION → pick question, answer (and password when changing) → SAVE	FR-19, FR-20
UC-09	Reset a forgotten password	Player	Login → FORGOT PASSWORD? → username or email → NEXT → answer, new password twice → RESET PASSWORD	FR-21 to FR-24
UC-10	Set up a lineup	Player	PLAY → pick department → pick 3 units → NEXT	FR-25 to FR-28
UC-11	Play a story stage	Player, AI	Pick a stage → FIGHT! → battle → results	FR-29 to FR-41, FR-44
UC-12	Play Endless mode	Player, AI	Stage Select → ENDLESS → survive → results	FR-43
UC-13	Buy a tower upgrade	Player	UPGRADES → choose department → spend coins	FR-42
UC-14	Host or join LAN 1v1	Player, second player	LAN 1V1 → HOST GAME or type IP → JOIN GAME → battle → results	FR-45 to FR-51
UC-15	Change database settings	Player, MySQL server	Login → DATABASE → enter server details or THIS COMPUTER → CONNECT	FR-52 to FR-55
UC-16	Change options	Player	OPTIONS → toggle settings	FR-56, FR-57

Example: UC-09 Reset a forgotten password

- **Precondition:** the account has a security question.
- **Main flow:** 1) Player presses FORGOT PASSWORD? on the Login screen. 2) Player types their username or email and presses NEXT. 3) The system shows the account's security question. 4) Player types the answer and the new password twice, then presses RESET PASSWORD. 5) The system checks the answer, saves the new password hash with a new salt, and shows "PASSWORD CHANGED". 6) Player returns to Login with the username filled in.
- **Alternative flows:** no such account → "NO ACCOUNT WITH THAT USERNAME OR EMAIL"; no question set → "THIS ACCOUNT HAS NO SECURITY QUESTION YET"; wrong answer → "WRONG ANSWER (n TRIES LEFT)"; fifth wrong answer → reset is locked for 15 minutes. Nothing is changed in these cases.
- **Postcondition:** only the new password works at the next login, and the wrong-answer counter is back to 0.

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